For the first year of my Masters in Intelligent Information Systems at Carnegie Mellon University (CMU), I completed my studies from my family's house. I juggled two new roles here: graduate student and caregiver for my older brother *[redacted]*. *[redacted]* has Duchenne Muscular Dystrophy, a degenerative neuromuscular disease which, at his age, confines him to a wheelchair and makes him physically unable to do most things that used to bring him joy. At home, I noticed he uses virtual assistants, such as Google Home, to carry out many functions he used to do manually: discovering information, organizing his life, and learning languages. They help him to enrich his life without relying on other humans for assistance.

I began working with *Prof. A* and *Dr. B* at CMU on information extraction ("IE"), a field of natural language processing aiming to turn text into structured knowledge, often as knowledge graphs. I am now applying for the PhD program in **<degree name>** at **<University>** in preparation for a career as a professor. My research goal is to improve the ability for people to easily access useful knowledge in unstructured text, with an eye towards important applications like scientific literature understanding and web accessibility.

Information extraction is an extensively studied subfield of NLP. Despite this, users must still pick between manually constructed knowledge bases (with low coverage) and automatically constructed knowledge bases (typically with low precision). These knowledge bases contain static information about the world, which can be used by humans or computers. Incorporating structured knowledge into downstream tasks can help interpretability, modularity, and reasoning, but is the predominant paradigm of using static, discrete knowledge bases still the best framework for delivering these objectives?

After leading a project at CMU on extracting information from scientific papers, I started thinking about the role that IE as a discipline plays within the broader utility of NLP for society. In our project, we made gains on standard metrics (such as *F1 score* calculated by treating IE as classification), leading to a paper at the *ACL 2021* conference [1]. Yet, I wondered if our work would make a tangible difference to downstream users. This question was validated after an ACL keynote talk by Christopher Potts, who urged the community to consider the broad consequences of models, beyond improving F1 metrics [2].

Building on my previous research on information extraction, I intend to go beyond static knowledge graph ("KG") construction and work towards systems that can jointly represent and reason about knowledge to ultimately improve natural language interfaces that broaden people's access to knowledge.

1) IE Evaluation: In recent years, machine translation researchers have developed new metrics that correlate better with human judgments than BLEU. I want to take a similar approach for IE, by measuring how well IE systems extract task-oriented knowledge for downstream tasks and designing intrinsic metrics that correlate with these extrinsic benchmarks better than today's *F1*-style metrics do.

2) Generating Task-Specific Reasoning Graphs: Knowledge graphs can help describe the "reasoning path" of a model to human decision-makers. Without relying on a knowledge graph, can we directly generate a reasoning path about entities in the environment from text while solving a task?

3) Improving Virtual Knowledge Bases: Recent work has studied "virtual" KGs [3], where edges between entities are not discrete but parameterized by a neural network. This approach enables multi-hop reasoning over raw text but lacks the interpretability and modularity of traditional KGs. Can we make the knowledge learned for one task more interpretable, controllable, and transferable to another related task?

4) **Making Knowledge More Accessible:** The growth of self-supervised models that encode semantics opens the possibility of representing heterogeneous data sources from different domains or modalities in one place. Building on the prior point, can we make knowledge from multiple ontologies, KGs, and text corpora accessible from a single natural language interface? This could give *[redacted]* access to specialized, multimodal knowledge, beyond the general factoids he is limited to with today's digital assistants.

I have seen the importance of proper evaluation in the space of spoken language understanding (SLU). In work published at *Interspeech 2021*, we found a shortcoming in a popular SLU dataset that we corrected via a principled evaluation method reflecting more realistic usage scenarios [4]. I came up with one of the central ideas of our paper, using multiple non-random test splits to test different characteristics of an SLU system. I envision that this spirit of designing evaluation procedures with the downstream usage of systems in mind can improve evaluation in complex semantic tasks like information extraction.

The idea of using knowledge graphs to aid reasoning in other tasks is related to the idea of exploiting problem structure to give better solutions. In past work, I have repeatedly seen value in this strategy. For our *ACL 2021* work, we used the citation structure in scientific documents to improve scientific IE after I noticed via error analysis that neural language models were making local decisions, ignoring global context. I further pursued this direction during an internship with *Professor C* at the Allen Institute for AI (work ongoing) where we are developing task-specific pretraining schemes to improve relation extraction, inspired by challenges found in a new biomedical IE dataset that I helped build [5].

I am grateful to have resumed my academic career during my Master's at CMU. In college, from 2012 to 2016, I struggled personally and did not fulfill my potential in school. During four years working on machine learning in industry after college, I regained confidence in myself and found a desire to work on problems with broader impact and visibility. In my Master's, I've managed multiple research projects while improving my foundations with courses in theory (like Probabilistic Graphical Models and Convex Optimization) and application (like Question Answering and Multilingual NLP). I showed myself I can thrive in an academic environment and enjoy doing research full-time. After seeing parallels between my work and the tools *[redacted]* uses, I am motivated to do research that improves access to information for all.

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[1] **Vijay Viswanathan**, Graham Neubig, Pengfei Liu. CitationIE: Leveraging the Citation Graph for Scientific Information Extraction. In *ACL-IJCNLP*, 2021.
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